MINING DEALER BEST PRACTICE

# Backpack for Field Service Electricians Tools

**Maintenance and Repair** | Application | Component Life Management | Component Renewal (CRC) | MARC Management






**October 17**
**1.04-161016-01**

## 1.0 Introduction

The field service technicians, deployed to diagnose and potentially resolve unscheduled stops of the equipment, are the first “line of defense” that the maintenance organization has to properly manage these failures, and ensure the minimum interruption to the scheduled work.

The "paramedic" analogy is often used to graph the role and responsibilities of field technicians, which have to respond quickly and arrive to the emergency location well prepared to provide an effective solution.

The field service is by nature a dynamic function that requires from the field personnel 4 major features: Response - Diagnosis - Decision - Repair.

The response and proper diagnosis is the key to the two later stages - making the right decision and executing the needed repairs. It is important to remember that depending on the decision taken the repair may or may not be performed in the field.

Proper response involves arriving at the right time (response time) and arriving well equipped. Arriving to the machine well equipped, means that the technician comes to service the equipment with all the appropriate tools to do the job.

This best practice presents a solution for ensuring the field electrical technicians are properly equipped by providing the technicians with a special backpack for carrying the needed tools to support their work. The backpack also improves their safety when accesses the machine.

## 2.0 Best Practice Description

This best practice is very simple in terms of its concept and implementation, and translates into better response times, diagnoses and personal safety.

Each leading field technician was equipped with a "Proto" backpack, which was equipped with the recommended and most used tools (electrical type of tools) for the type of work to be performed, in this case, diagnosis and repair of electrical faults.



Technician Carrying a “Proto” Electric Backpack

As previously mentioned the main objectives of this BP are to improve the equipment of technicians, improve the response time and quality of their work and, most importantly, contribute to the safety of field service personnel.

In regards to safety, with the implementation of this proposed backpack, we free the hands of the technicians when climbing to the equipment enabling them to observe and always maintain 3 points of support (rule of the 3 points of support).

In addition, when the technician accesses the equipment with all the necessary equipment, it reduces the need for extra trips to get tools, improving the time of repair and safety by reducing trips made from the machine to the technician’s truck.

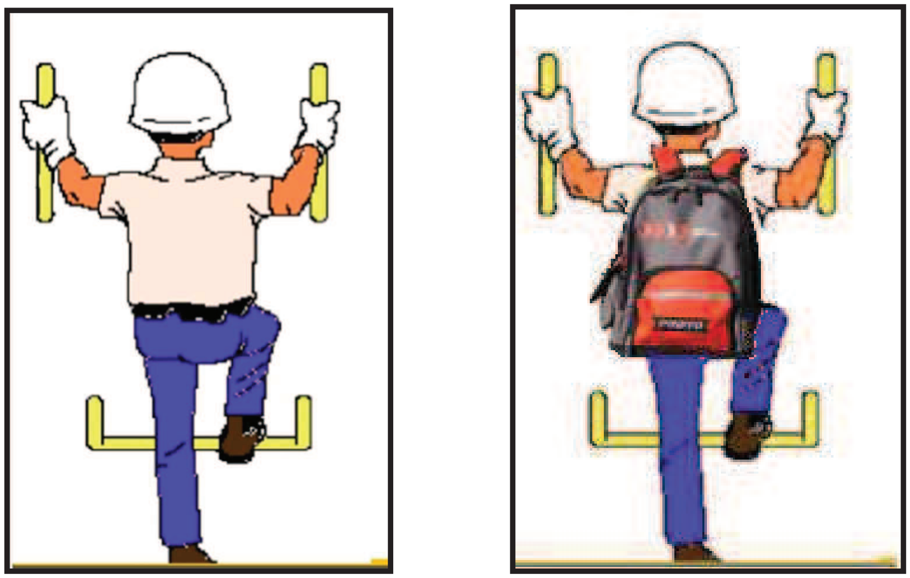

Diagram illustrating the rule of 3 supports, when climbing to an equipment. Note the schematic on the right where the technician is equipped with his Proto backpack for power tools.

Photograph of the backpack Proto equipped with the appropriate tools to perform tasks of the electrical type.

## 3.0 Implementation Steps

The general steps we recommend to follow for the implementation of this Best Practice are:

**3.1 Selection of the Backpack to use**
**3.2 Define Tool list**
**3.3 Practical Tests**
**3.4 Final Adjustments**
**3.5 Equip your Service Field Technicians crew**
**3.6 Document the SOP (Standard Operating Procedure)**
**3.7 Training**
**3.8 Operational Start-up and Benefit Validation**

### 3.1 Selection of the Backpack to use

In our case we selected the backpack of the Proto brand, model J114BP, which met our requirements.

| Brand | Model | Description | Photo |
|---|---|---|---|
| **PROTO (USA)** | **J114BP** | **BACKPACK W/ REMOVABLE TOTE**<br>Durable 1,200 x 1,200 denier fabric.<br>Rigid and waterproof plastic bottom.<br>Internal storage compartments for maximum organization<br>Convenient laptop storage included<br>Quick carry tool pouch included for easy transport of commonly used items<br>Height 18 Inches, Depth 11 inches<br>Weight 4.5 pounds |  |

The characteristics we looked at at the time of selection were:

- Durability, this backpack is made with a fabric of 1200 x 1200 denier
- Rigid waterproof plastic bottom
- Internal storage compartments for maximum organization
- Convenient laptop storage included
- Quick carry tool pouch included for easy transport of commonly used items

Convenient laptop storage included



Quick carry tool pouch included for easy transport of commonly used items

Rigid and waterproof plastic bottom

## 3.2 Define Tool list

Careful selection of the tools with which the backpack is to be equipped is important. This will depend on the type of equipment that must be serviced in the field and, the past experience with the type of failures that the equipment normally presents.

In our case at Antucoya we serve a fleet of 11 Caterpillar model 797 A trucks, and our choice of tools was made in conjunction with the field service technicians.

The following is the list of tools and test equipment we have selected and currently using.

| List of Tools in Backpack | | | |
|---|---|---|---|
| ***Ítem*** | ***Description*** | ***Part Number*** | ***Quantity*** |
| 1 | Wire Stripper | 1U5804 | 1 |
| 2 | Circuit Tester | 8T0500 | 1 |
| 3 | Voltmeter | 5P7277 | 1 |
| 4 | Terminal Kit | 1409944 | 1 |
| 5 | Tips | 8T3224 | 1 |
| 6 | Multimeter | 2375130 | 1 |
| 7 | Cable | 3264904 | 1 |
| 8 | Cable Probe Group | 7X1710 | 1 |
| 9 | Removal Tool | 1219588 | Q |
| 10 | Removal Tool | 1516320 | 10 |
| 11 | Removal Tool | 1957970 | 10 |
| 12 | Plier Set | 2331639 | 1 |
| 13 | Wrench 3MMHX | 1U7300 | 1 |
| 14 | Screwdriver Group | 1627804 | 1 |
| 15 | Seal Pick | 5P7414 | 1 |
| 16 | Tape Measure | 5P3277 | 1 |
| 17 | Key Set Hex MM | 2001931 | 1 |
| 18 | Key Set Hex IN | SN | 1 |
| 19 | WRENCH-TEEHA 5/32 | 1730960 | 1 |
| 20 | Pick Up TL | 1462728 | 1 |
| 21 | Socket Set MM | 1943574 | 1 |
| 22 | Joint-Universal 3/8 | 6V0094 | 1 |
| 23 | Ratchet 3/8 | 2396824 | 1 |
| 24 | Extension-Short | 8H8574 | 1 |
| 25 | Extension-Long | 8H8575 | 1 |
| 26 | Socket 7/16 | 8H8567 | 1 |
| 27 | Socket 9/16 | 8H8569 | 1 |
| 28 | Socket 1/2 | 8H8568 | 1 |
| 29 | Socket 5/8 | 8H8570 | 1 |
| 30 | Socket 3/4 | 8H8571 | 1 |
| 31 | Socket 11/16 | 9S1728 | 1 |

A laptop, spare computer battery, or charger for use in the field and a radio are also important items to add to the bags contents.

### 3.3 Practical tests

Before placing the backpacks in operation, we recommend testing one or two of them with the field electricians. This is to assess whether some adjustment needs to be made primarily to the tool content.

### 3.4 Final Adjustments

At this stage, adjustments should be made as necessary.

### 3.5 Equip your Service Field Technicians crew

It is important to equip the field crews with backpacks containing the appropriate quantity and types of tools, to ensure minimum waiting time for tools in the field tasks.

In our case we decided to assign four (4) backpacks per field service shift.

### 3.6 Document the SOP (Standard Operating Procedure)

Local specific safety regulations should be reviewed, in order to adapt the standard operating procedures to your specific needs and/or regulations. The SOP should be documented to insure the backpacks are properly used and maintained.

### 3.7 Training

It is important to properly train the personnel who will use backpacks routinely to ensure its correct use and maintenance.

We recommend that this training includes the use and care of the tools and backpack assigned to the field staff.

### 3.8 Operational Start-up and Benefit Validation

Once the backpacks are in operation, the correct use and benefits should be tracked and recorded.

It is important to observe the response time results of the field staff as well as the time of diagnosis and repair of this type of tasks.

An important point when putting this backpack into service is to implement a system for auditing the content and conditions of the assigned tools and backpacks. This is important to ensure the proper operation of this equipment over time.

## 4.0 Benefits

As mentioned above the main benefits of this best practice are located in: Safety, Response Time, Diagnostic and Repair Time, and Quality of Repairs.

**Safety:** The use of the recommended backpack ensures the proper transport of the tools allowing the technician to respect the rule of the three supports when climbing the equipment. It also eliminates unnecessary trips to search for missing tools.

**Response Time:** Response time is significantly improved by having properly equipped technicians. Technicians come to the machine with all the tools they need.

The improvements in response time have a positive effect on the MTTR results of the fleet.

**Diagnostic and Repair Time:** As with the response time, having technicians fully equipped with the correct tools, improves the time of Diagnosis and Repair.

Diagnostic and Repair time improvements have a positive influence on fleet MTTR results.

**Quality of the Repairs**: An accurate diagnosis helps to make an effective repair. In addition field repairs performed using the appropriate test tools and equipment will be of better quality.

Improvements in repair quality have a positive effect on the MTBS and MTBF results of the fleet.

## 5.0 Resources Required

The required resources are those specified in section three (3) of this BP.

Note: The published cost for the selected backpack is approximately US $ 80.

## 6.0 Supporting Attachments / References

**N/A**

## 7.0 Related Best Practices

**N/A**

## 8.0 Acknowledgements

Generation and Implementation of Best Practice

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